

Module 3: Product/Market Fit & Customer Discovery Questions

Instructions:

1. After testing your hypotheses through customer interviews, answer the questions in the chart below about your validated segments concisely and completely within the character limits (**5,000 characters MAX**).

Product/Market Fit & Customer Discovery Questions: (5,000 character limit total)

Questions	Answers
How many customer interviews have you done since the beginning of the CTO Accelerator?	14 customer discovery interviews with commercial building owners, property managers, facilities directors, real estate investment trusts (REITs), retail franchisees, sustainability heads, and energy consultants across Colorado and nationally.
Specify the validated market segments you have identified, in which the customers buy similar products, have a similar sales cycle, look for a similar value proposition, and reference each other.	Our primary validated segment is mid-large sized apartment buildings (~50,000 sq ft). These sites share common traits: outsourced facilities management, rising energy costs, fragmented legacy systems, and increasing pressure to manage site performance and capital improvements. Buyers and influencers include asset managers, property managers, and heads of operations.
Identify specifically the pain your technology addresses/problem it solves: a monetizable, "shark bite," "hair on fire" pain or problem validated by your interviews.	Buildings are black boxes when it comes to energy: operators have no visibility into where energy is wasted, no way to control systems dynamically, and no ability to optimize across devices or respond to grid signals. This leads to chronic overuse, inflated utility costs, poor capital planning, and missed revenue opportunities. The pain is urgent as operating costs rise, tenants demand sustainable practices, and infrastructure upgrades loom.
In your value proposition, describe your product briefly and how it addresses that pain or problem, excites the customer, taps an urgent need, and/or will have a big impact on	SPARK's Intelligent Energy Hub (IEH) is a plug-and-play AI device that installs onto a building's electrical panel and transforms it into an intelligent energy node. It gives

their business. Quantify the impact, if possible. (This value proposition should get prospective customers to return your cold call).	businesses real-time insight, intelligent coordination, and direct control over how energy is used—at the circuit, system, and site level. It turns wasted energy into strategic savings. In interviews, customers reported willingness to pay for 8–15% savings on energy bills (~\$1.50–\$3.00/sq ft annually), faster ROI on electrification upgrades, and the ability to offer load flexibility to aggregators/utilities.
Describe the minimum feature set that your customer has clearly indicated in interviews they want more than other features. What will they pay for this Minimum Viable Product as the first release?	The MVP includes (1) real-time circuit-level monitoring, (2) device-level disaggregation and system mapping, (3) a dashboard with alerts and usage breakdowns, and (4) automated intelligent scheduling of HVAC and lighting. Control integration is optional at first but highly requested for the next phase. Customers indicated willingness to pay \$6,000–\$8,000 upfront for installation and ~\$250–\$500/month in software fees per building, with pricing aligned to building size and savings delivered.
What are the key risks to produce just your MVP, whether financial (cost to develop) or technological (new invention or process), etc?	Key risks include: (1) hardware integration complexity—reliability of sensors, electrical code compliance, installation variability; (2) achieving high-resolution, accurate disaggregation through AI models; and (3) UI/UX performance—ensuring actionable insights and usability. Financial risk centers on hardware prototyping, pilot deployment, and iterative development.
What's your revenue model? Will you (a) sell a product or service or lease or (b) sell subscriptions or (c) license the technology or product?	We offer a hybrid model: (a) one-time hardware sale (device + install), (b) recurring monthly SaaS fee for insights, scheduling, reporting, and control, and (c) potential revenue share with utilities for

	grid-interactive services once functionality is proven.
How will you price the product? Does your pricing model take into account competitive pricing, manufacturing costs, product value, and “how much the market will bear?”	Hardware pricing is designed to be under \$10,000 per building for a system supporting 100–200 circuits. SaaS pricing is based on savings value and market comparable (e.g., building management systems, monitoring services, controls platforms). Our model aims for a 12–18 month payback period for the customer.
Are you creating a new market, or have you found a niche in an existing market that you can dominate? If you’re in a well-established market, what is your competitive advantage in your subsegment that drives a customer’s purchase decision and against entrenched competitors or suppliers of an alternative solution?	We’re targeting a neglected segment of an existing market: mid-sized buildings that fall below the radar of traditional energy service companies and don’t have the budget or complexity for full building management systems. SPARK provides an “intelligence layer” where there’s currently nothing—no data, no coordination, no interface. Our competitive edge is plug-and-play intelligence, AI-driven decision support, and progressive upgradeability without major retrofits.
Estimate the size of your target customer sub-segment (NOT THE TAM OR SAM) and calculate the \$ size of that segment, using your validated revenue model and pricing structure.	We estimate ~70,000 qualifying commercial buildings in the U.S. that fall into our core sub-segment. At an average annual revenue of \$8,000–\$10,000 per building (hardware + SaaS), this segment represents an \$500M opportunity, independent of DER integrations or grid revenue.
Provide some evidence or insights that show whether or not your target sub-segment represents your first million-dollar market within a year.	Based on interviews, we estimate pilot conversion and upsell within ~100 buildings in Year 1. With an average revenue of ~\$15K per building over the first 12–18 months (hardware + SaaS + optional services), that totals over \$1M in Year 1 revenue potential. Additional value

	could come from grid participation as DER orchestration scales.
If you win this sub-segment, describe how you can leverage it to enter adjacent markets with only slight modification to the product or your sales strategy.	Success in this segment builds a replicable sales and installation model for other asset-heavy sectors: strip malls, retail chains, grocery stores, schools and municipal buildings. The same IEH hardware and software can scale horizontally—simply re-skinned with industry-specific dashboards, or configured for different circuit sizes and load profiles.
In the target segment, have you sold “early evangelists” who have bought into your vision and provide references/testimonials, or pilot customers who will collaborate with you to define the Minimum Viable Product?	We are working towards securing commitments for early pilots with two Denver-based commercial operators. These partners are currently involved in product feedback, prioritization of features, and co-developing a pilot roadmap.